

Context

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Experiment No: 01

Experiment Name: Design and implementation of a Star topology in Cisco Packet Tracer.

Objectives:

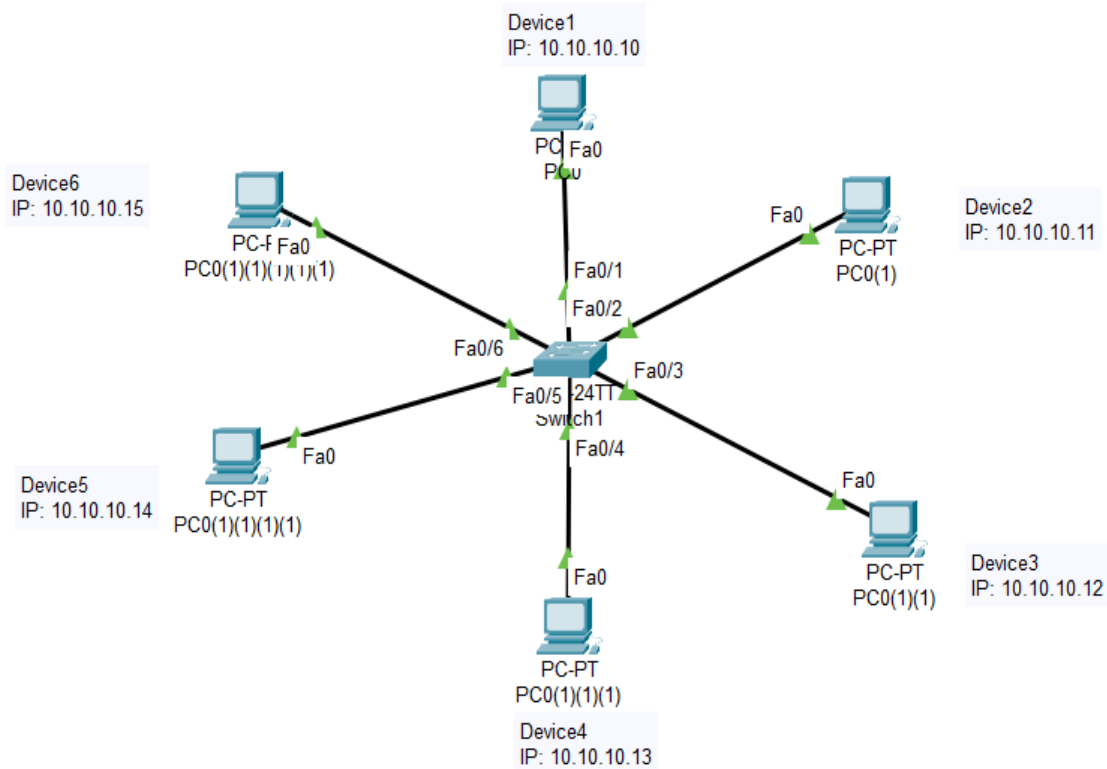
- ☐ Learned how to configure devices such as switches, routers, and end devices in a star topology.
- ☐ Established proper connections between devices and assigning IP addresses for communication.
- ☐ Verified end-to-end connectivity through simulation (e.g., using ping commands).
- ☐ Analyzed the benefits of using a star topology, such as ease of management, scalability, and fault isolation.

Requirements:

- ❖ 1 Switch (central device).
- ❖ Here used 6 PCs.
- ❖ Ethernet cables (Copper Straight-through).
- ❖ Cisco Packet Tracer software.
- ❖ IP addressing scheme for all PCs.
- ❖ Basic knowledge of assigning IP addresses to end devices.

Design:

1. Design and implementation of a Star topology in Cisco Packet Tracer.



Designed by Md.Mahfujul Karim Sheikh

Working Procedure:

1. Opened **Cisco Packet Tracer**.
2. Drag and drop **one switch** into the workspace.
3. Added **6 PCs** around the switch.
4. Connected each PC to the switch using **Straight-through cables**.
5. Assigned unique **IP addresses** to each PC.

6. Test connectivity using the **ping command** from one PC to another.
7. Verify that all devices can communicate through the central switch (Star Topology).

Result:

```
C:\>ping 10.10.10.15

Pinging 10.10.10.15 with 32 bytes of data:

Reply from 10.10.10.15: bytes=32 time<1ms TTL=128
Reply from 10.10.10.15: bytes=32 time<1ms TTL=128
Reply from 10.10.10.15: bytes=32 time<1ms TTL=128
Reply from 10.10.10.15: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Experiment No: 02

Experiment Name: Design and implementation of a Mesh topology in Cisco Packet Tracer.

Objectives:

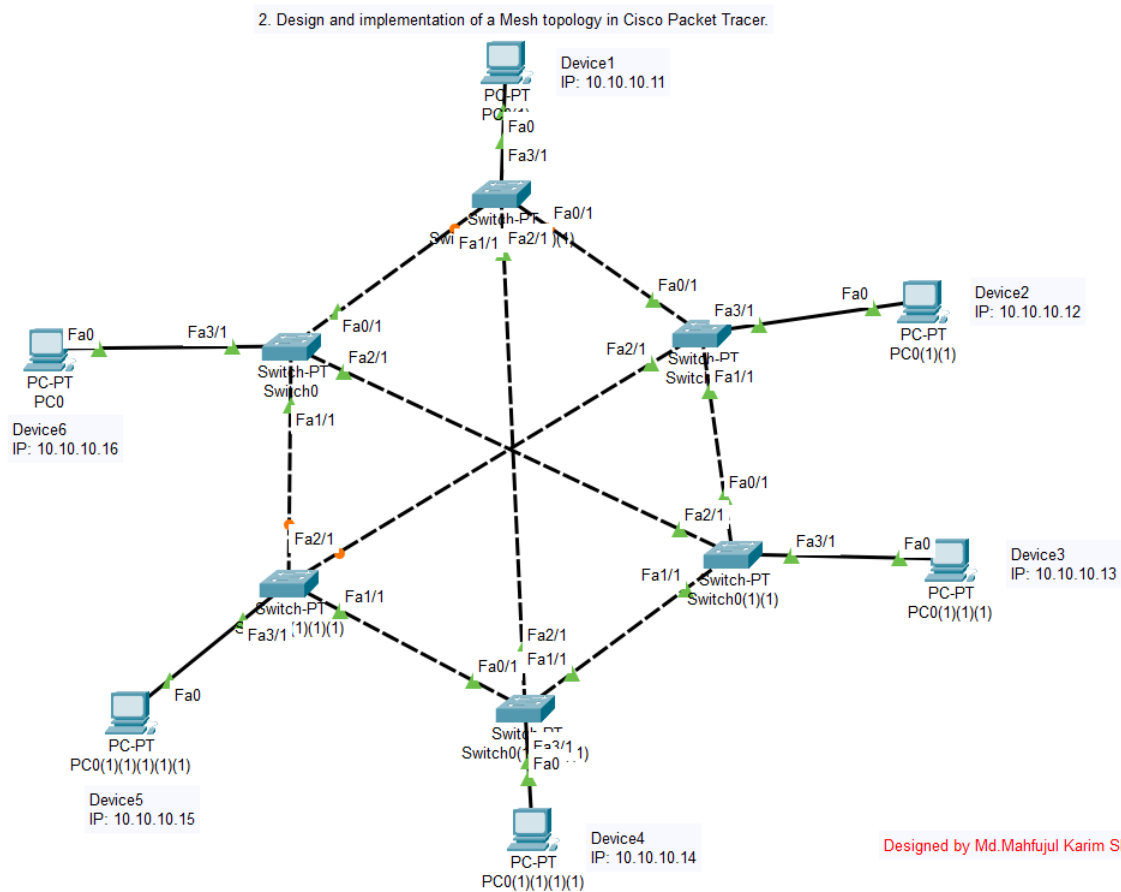
- ☐ Learned how to interconnect multiple devices so that each device has direct communication with others.
- ☐ Configured IP addressing for all connected devices.
- ☐ Verified data transfer reliability and fault tolerance using simulation tools.
- ☐ Analyzed the advantages of mesh topology in terms of redundancy, fault tolerance, and reliability.

Requirements:

- ❖ 6 PCs and 6 Switch and cables.
- ❖ Each PC connected to every other PC (point-to-point links).

- ❖ Ethernet cables (Copper Cross-over or Straight-through depending on device).
- ❖ Cisco Packet Tracer software.
- ❖ Proper IP addressing plan.
- ❖ Understanding of direct device-to-device communication.

Design:



Working Procedure:

1. Opened Cisco Packet Tracer.
2. Placed **6 PCs**, **6 switch** and **necessary cables** on the workspace.
3. Connected each PC to every other PC (partial mesh) using Ethernet cables.
4. Assigned a **unique IP address** to each PC.

5. Used the **ping command** to check connectivity between all pairs of PCs.
6. Verified that communication is successful without a central device (direct links).

Result:

```
C:\>ping 10.10.10.13

Pinging 10.10.10.13 with 32 bytes of data:

Reply from 10.10.10.13: bytes=32 time<1ms TTL=128
Reply from 10.10.10.13: bytes=32 time<1ms TTL=128
Reply from 10.10.10.13: bytes=32 time<1ms TTL=128
Reply from 10.10.10.13: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.10.13:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Experiment No: 03

Experiment Name: Static routing configuration for four interconnected networks.

Objectives:

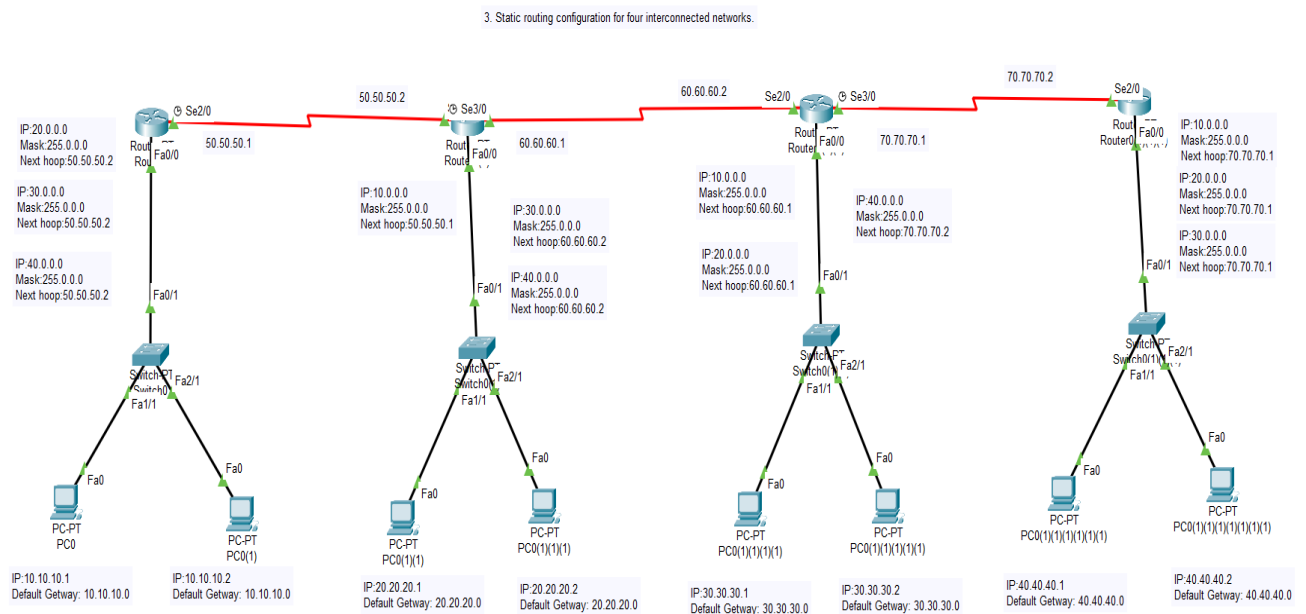
- ☐ Understood the concept of static routing and manual route entry.
- ☐ Assigned proper IP addressing schemes to all devices in the network.
- ☐ Configured static routes in routers to ensure inter-network communication.
- ☐ Tested connectivity between all networks using ping and traceroute.
- ☐ Analyzed the advantages and limitations of static routing in real-world scenarios.

reliability.

Requirements:

- 4 Routers (to represent 4 networks).
- 4 Switches (for connecting PCs).
- 8 PCs.
- Ethernet cables (Straight-through between PC ↔ Switch, and Cross-over between Router ↔ Router if required).
- IP addressing scheme (subnet planning).
- Cisco Packet Tracer software.
- Knowledge of `ip route` command for static routing configuration.

Design:



Designed by Md. Mahtujul Karim Sheikh

Working Procedure:

- Place **4 routers** in the workspace to represent four different networks.

- Connected each router to its own **switch and PC** .
- Interconnected the routers using serial or Ethernet cables.
- Assigned **IP addresses** to all PCs, router interfaces, and inter-router links.
- On each router, configured **static routes** using the ip route command so that each router knows how to reach other networks.
- Tested connectivity by pinging PCs from one network to another.
- Verified that routing is successful only because of the manually configured static routes.

Result:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 40.40.40.2

Pinging 40.40.40.2 with 32 bytes of data:

Request timed out.
Reply from 40.40.40.2: bytes=32 time=42ms TTL=124
Reply from 40.40.40.2: bytes=32 time=40ms TTL=124
Reply from 40.40.40.2: bytes=32 time=25ms TTL=124

Ping statistics for 40.40.40.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 42ms, Average = 35ms

C:\>ping 30.30.30.1

Pinging 30.30.30.1 with 32 bytes of data:

Request timed out.
Reply from 30.30.30.1: bytes=32 time=35ms TTL=125
Reply from 30.30.30.1: bytes=32 time=46ms TTL=125
Reply from 30.30.30.1: bytes=32 time=3ms TTL=125

Ping statistics for 30.30.30.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 46ms, Average = 28ms
```

Experiment No: 04

Experiment Name: Dynamic routing configuration for four interconnected networks..

Objectives:

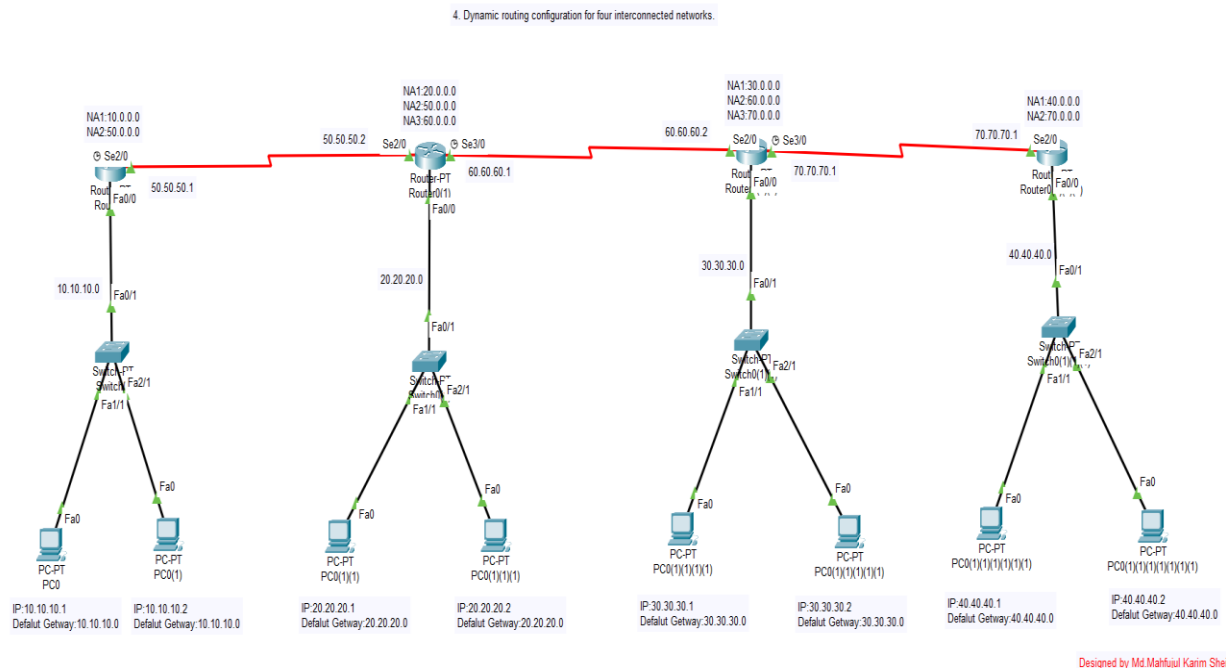
- ☐ Understood the working principles of dynamic routing protocols .
- ☐ Configured routers to automatically share routing information.
- ☐ Verified automatic route updates and communication between all networks.
- ☐ Compared dynamic routing with static routing in terms of scalability and efficiency.
- ☐ Analyzed the role of dynamic routing in large and complex networks.

Requirements:

- 4 Routers (to represent 4 different networks).
- 4 Switches (for end devices).
- 8 PCs.
- Ethernet cables (Straight-through & Cross-over).
- IP addressing scheme.
- Cisco Packet Tracer software.
- Knowledge of dynamic routing protocols (e.g., RIP, OSPF, or EIGRP).
- Commands for enabling and configuring routing protocols.

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Design:



Working Procedure:

- Placed **4 routers**, each connected to a **switch** and **PC**.
- Connected routers together using cables.
- Assigned IP addresses to PCs, router interfaces, and inter-router connections.
- Enabled a **dynamic routing protocol** (e.g., RIP, OSPF, or EIGRP) on all routers.
- Ensured all routers exchange routing information automatically.
- Tested connectivity using ping between PCs in different networks.
- Verified that communication is successful without manually configuring static routes.

Result:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=3ms TTL=124
Reply from 10.10.10.2: bytes=32 time=47ms TTL=124
Reply from 10.10.10.2: bytes=32 time=4ms TTL=124
Reply from 10.10.10.2: bytes=32 time=3ms TTL=124

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 47ms, Average = 14ms

C:\>ping 20.20.20.1

Pinging 20.20.20.1 with 32 bytes of data:

Request timed out.
Reply from 20.20.20.1: bytes=32 time=26ms TTL=125
Reply from 20.20.20.1: bytes=32 time=2ms TTL=125
Reply from 20.20.20.1: bytes=32 time=5ms TTL=125

Ping statistics for 20.20.20.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 26ms, Average = 11ms
```

Experiment No: 05

Experiment Name: Design of a complete campus network.

Objectives:

- ☐ Designed a hierarchical network structure with core, distribution, and access layers.
- ☐ Implemented VLANs, IP addressing, and inter-VLAN routing.

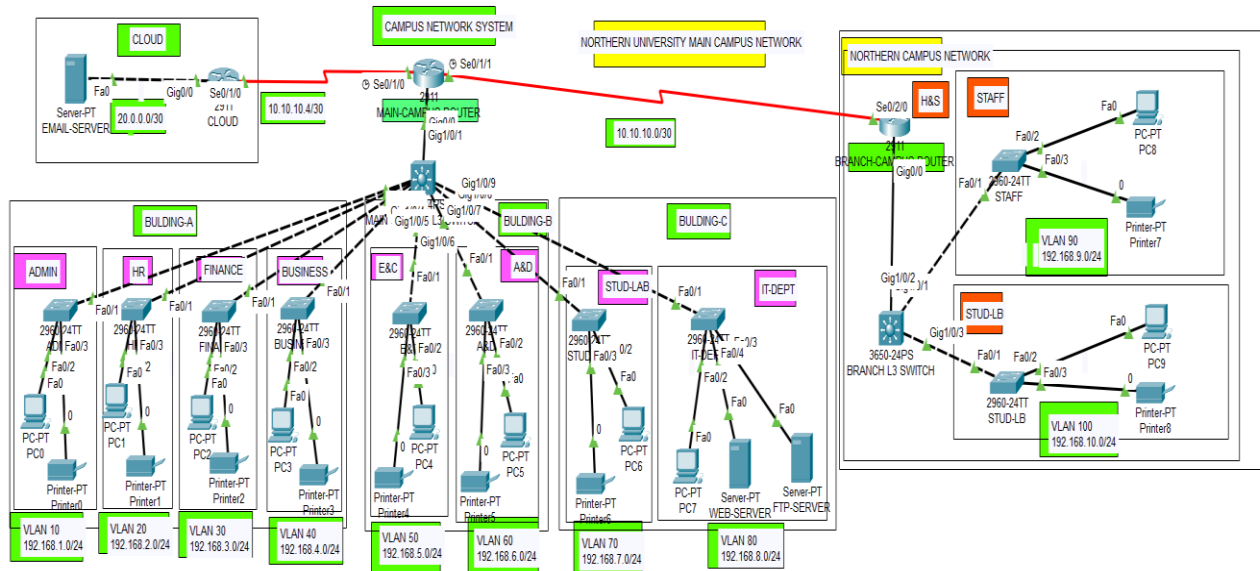
- ☐ Configured essential services such as DHCP, DNS, and server-based applications.
- ☐ Ensured secure and efficient communication between different segments of the campus network.
- ☐ Analyzed the performance, scalability, and security aspects of a campus network design.

Requirements:

- Multiple Routers (core, distribution, and edge).
- Multiple Switches (access and distribution layer).
- PCs/Laptops (for students, staff, admin, etc.).
- Servers (DNS, DHCP, Web, Email).
- Wireless Devices (Access Points, Wireless Routers ☐ if needed).
- Proper IP addressing (subnetting, VLANs if required).
- Cisco Packet Tracer software.
- Knowledge of:
 - VLANs and Inter-VLAN routing.
 - Static & Dynamic routing.
 - DHCP server configuration.
 - Basic security (Access Control Lists).

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Design:



Working Procedure:

- Opened Cisco Packet Tracer and plan a **hierarchical network design** (Core □ Distribution □ Access).
- Placed **routers** (for backbone), **switches** (for different departments/labs), and **PCs/servers**.
- Connected end devices to access switches, access switches to distribution switches, and then to the core router.
- Assigned IP addresses using **subnetting** to separate departments (e.g., Students, Faculty, Admin).
- Configured **VLANs** for departmental segmentation and use **Inter-VLAN routing**.
- Added a **DHCP server** to assign IPs automatically, and configure **DNS/Web/Email servers**.

- Used a **dynamic routing protocol** (like OSPF) for inter-router communication.
- Tested connectivity using ping and traceroute across different parts of the network.
- Verified that the complete campus network is functional with internet, server access, and inter-department communication.

Result:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 169.254.190.203

Pinging 169.254.190.203 with 32 bytes of data:

Reply from 169.254.190.203: bytes=32 time=1ms TTL=128
Reply from 169.254.190.203: bytes=32 time<1ms TTL=128
Reply from 169.254.190.203: bytes=32 time<1ms TTL=128
Reply from 169.254.190.203: bytes=32 time=3ms TTL=128

Ping statistics for 169.254.190.203:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

NB: For implementing campus area network , I used directly online resources and AI. Source is collected from this site: https://github.com/Anas436/Campus-Network-System-Using-Cisco-Packet-Tracer/blob/main/Final_Project_CampusNetwork.pkt .